

End of Year Project Report Synthesis Project II

Conception of Reinforcement learning-based Robotic Arm

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Dédications

Je voudrais dédier ce rapport à mes parents pour leur soutien indéfectible tout au long de mon parcours académique, qui ont été une source constante d'inspiration, de valeurs éthiques, de persévérance et de détermination. Leur soutien moral et financier m'a permis d'atteindre cette étape académique. J'aimerais également dédier ce rapport à mon superviseur, M. Zied Antar, ainsi qu'à Madame Houda Hachem, qui a accepté de faire partie du jury pour évaluer ce travail. Enfin, je souhaite exprimer ma gratitude à mes amis pour leur soutien et leur confiance. Je remercie également l'École Nationale des Sciences et Technologies Avancées de Borj Cedria pour avoir rendu cette opportunité possible.

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LIST OF ABBREVIATION

AI - Artificial Intelligence

ARM - Advanced RISC Machine

CAD - Computer-Aided Design

CATIA - Computer-Aided Three-dimensional Interactive Application

CNN - Convolutional Neural Network

CPU - Central Processing Unit

DQN - Deep Q-Learning Networks

GHz - Gigahertz

GPU - Graphics Processing Unit

GUI - Graphical User Interface

HDMI - High-Definition Multimedia Interface

IDE - Integrated Development Environment

IoT - Internet of Things

LAN - Local Area Network

LSTM - Long Short-Term Memory

LAN - Local Area Network

LPDDR2 SDRAM - Low Power Double Data Rate 2 Synchronous Dynamic Random-Access Memory

LISP - List Processing

MicroSD - Micro Secure Digital

MDPs - Markov Decision Processes

NLP - Natural Language Processing

PCA - Principal Component Analysis

PLA- Polylactic Acid

PLM - Product Lifecycle Management

OS- Operating System

RNN - Recurrent Neural Network

RL - Reinforcement Learning

RGB - Red Green Blue

SD- Secure Digital

SDRAM- Synchronous Dynamic Random-Access Memory

SSH- Secure Shell

STL-stereolithography

SVM - Support Vector Machines

USB - Universal Serial Bus

WLAN - Wireless Local Area Network

GENERAL INTRODUCTION

In a world where technological advances are rapidly transforming the way we work, live and interact with our environment, artificial intelligence (AI) is emerging as one of the driving forces behind this transformation. AI, with its ability to learn from data, recognize complex patterns and make autonomous decisions, offers immense potential for solving complex problems in a variety of fields.

Robotic arm control is one such area where AI can revolutionize our ability to automate precise and sophisticated tasks. Robotic arms, with their ability to perform a wide variety of movements and actions, have applications in many sectors, from manufacturing to medicine to space exploration. However, precise control of these robotic arms in dynamic and unpredictable environments remains a major challenge.

In this context, this final year project report explores the use of reinforcement learning (RL) to improve robotic arm control. Reinforcement learning, a branch of AI that draws on the behavior of living beings to learn how to make optimal decisions in complex environments, offers a promising approach to training robotic arms to perform specific tasks autonomously.

This report presents an in-depth study of the design, construction and training of a robotic arm using reinforcement learning. We examine the various components of the project, from modeling the problem to implementing the RL algorithm on a Raspberry Pi to control the movements of the robotic arm. In addition, we evaluate the system's performance and discuss the challenges encountered throughout the development process.

Through this report, we aspire to contribute to the understanding and advancement of robotic arm control by exploring the possibilities offered by reinforcement learning. We hope that this study will serve as a basis for future research and innovation in this exciting and constantly evolving field.

Our report is structured into several parts to provide a comprehensive exploration of the topic. Chapter 1 provides an Introduction to AI and Reinforcement Learning (RL), laying the groundwork for understanding the underlying principles and technologies driving advancements in robotic systems. Building upon this foundation, Chapter 2 delves into the functionalities and applications of Robotic Arms, exploring their diverse uses across different industries and highlighting key features and capabilities. Finally, in Chapter 3, we focus on the Design and Implementation of a Reinforcement Learning-based Robotic Arm, showcasing the practical application of AI techniques in robotic control and autonomy.

This report presents an in-depth study of the design, construction, and training of a robotic arm using reinforcement learning. We examine the various components of the project, from modeling the problem to implementing the RL algorithm on a Raspberry Pi to control the movements of the robotic arm. In addition, we evaluate the system's performance and discuss the challenges encountered throughout the development process.

Through this report, we aspire to contribute to the understanding and advancement of robotic arm control by exploring the possibilities offered by reinforcement learning. We hope that this study will serve as a basis for future research and innovation in this exciting and constantly evolving field.

Chapter 1: Artificial Intelligence and Reinforcement Learning

Introduction:

In this introductory chapter, we will delve into the fundamentals of artificial intelligence (AI) and reinforcement learning (RL), thus laying the necessary groundwork to understand their application in robotic arm control.

1. Why We Chose This Project?

We chose to work on a robotic arm with reinforcement learning because we believe it holds great promise for transforming automation. By integrating reinforcement learning, we aim to make robots capable of learning and adapting on their own, which could greatly improve their performance in various situations. This project reflects our dedication to pushing the boundaries of robotics and using AI to solve practical problems.

2. Concept of Artificial Intelligence [1]

Artificial Intelligence (AI) is a multifaceted field comprising diverse techniques and methodologies. In this section, we will explore the theoretical underpinnings of Reinforcement Learning (RL), a subset of AI with significant implications for robotics and various other domains.

2.1. Definition of Artificial Intelligence

Artificial intelligence (AI) is a diverse field encompassing computer systems capable of tasks like problem-solving and decision-making, traditionally associated with human cognition.

In today's world, AI includes various technologies integrated into daily life, from personalized entertainment recommendations to instant customer support via chatbots.

Fundamentally, AI involves creating computer systems to imitate human-like intelligence, using disciplines like machine learning, deep learning, and natural language processing (NLP).

Debates question whether contemporary AI truly embodies artificial intelligence or is advanced machine learning.

The term "AI" generally refers to a spectrum of machine learning-driven technologies in modern applications, such as Chat GPT and computer vision systems, enabling tasks once thought exclusive to humans.

2.2. Historical Evolution of Artificial Intelligence

AI has undergone substantial alterations throughout the decades (Figure 1), moving from rule-based systems to more sophisticated techniques, encouraged by technical advancements: Warren McCulloch and Walter Pitts created a revolutionary artificial neuron model in 1943, laying the groundwork for artificial intelligence. Alan Turing's 1950 article, "Computing machinery and intelligence," prompted debates on machine cognition. The invention of the LISP programming language by John McCarthy in 1956 was a watershed moment. The 1990s saw an increase in processing power and datasets, which propelled AI into modern applications. In recent years, alongside these advancements, reinforcement learning (RL) has emerged as a powerful paradigm, transforming AI by allowing machines to learn and adapt from experience.

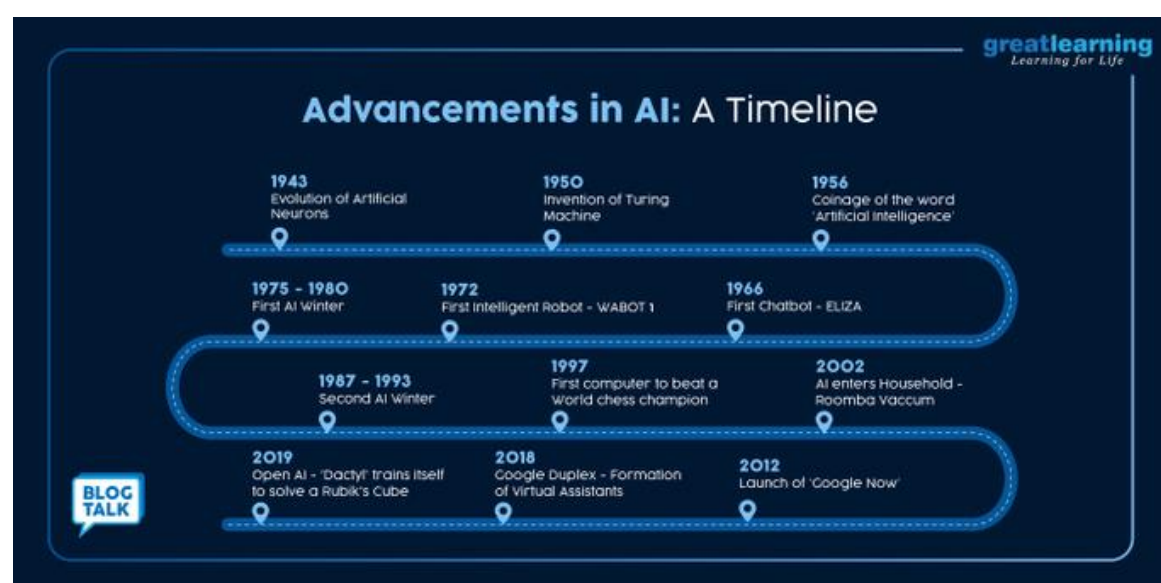


Figure 1: Historical Evolution of Artificial Intelligence

2.3. Main Fields of Artificial Intelligence [2]

AI encompasses Machine Learning, Natural Language Processing, Computer Vision, Robotics, Expert Systems, and Neural Networks, each contributing unique capabilities such as learning from data, understanding language, interpreting images, automating tasks. In this section, we are going to mention some of the most important fields of AI.

2.3.1. Machine Learning

Machine Learning is a prominent field in today's technology sphere, enabling computers to learn autonomously and seamlessly integrate into daily life. Its algorithms are developed on sophisticated mathematical concepts, enabling tasks like data categorization and interpretation from diverse datasets.

Recent advancements in Machine Learning have revolutionized various domains, introducing innovations like self-driving cars and image recognition. Actually, machine learning algorithms are typically categorized into three main types.

a. Supervised Learning

Supervised Learning is the process of instructing a model using labeled data, where inputs are matched with corresponding outputs. The goal is to create a map from inputs to outputs, allowing the model to predict or classify new, unseen data.

Common applications include image recognition and language translation. Let us mention some examples of algorithms used in supervised learning:

- **Decision Trees:** A classification algorithm that segments data into hierarchical structures based on features.
- **Linear Regression:** A regression algorithm that models the relationship between independent and dependent variables using a linear equation.
- **Support Vector Machines (SVM):** A classification algorithm that separates data points using a hyperplane with the maximum margin.

b. Unsupervised Learning

Unsupervised Learning involves instructing models using unlabeled data, devoid of explicit guidance or predefined outputs.

The aim is to unearth inherent patterns, structures, or relationships within the data.

Typical methodologies encompass clustering, dimensionality reduction, and anomaly detection, aiding in the revelation of latent insights and groupings in datasets without the necessity of labeled instances. Let us mention some examples of algorithms used in unsupervised learning:

- **K-Means Clustering:** An algorithm that partitions data into 'k' clusters based on similarity of features.
- **Principal Component Analysis (PCA):** A technique that reduces the dimensionality of data while preserving its variance, commonly used for feature extraction and data visualization.

c. Reinforcement Learning

Reinforcement Learning (RL) trains agents to make decisions by interacting with the environment (Figure 2) and receiving feedback as rewards or penalties. Unlike supervised learning, where actions are provided, RL agents learn through trial and error, exploring actions to optimize long-term rewards. This method is useful when the best action is uncertain or traditional programming is impractical. RL finds application in training autonomous vehicles and teaching robotic arms complex tasks. Through iterative learning, RL agents improve decision-making in dynamic environments.

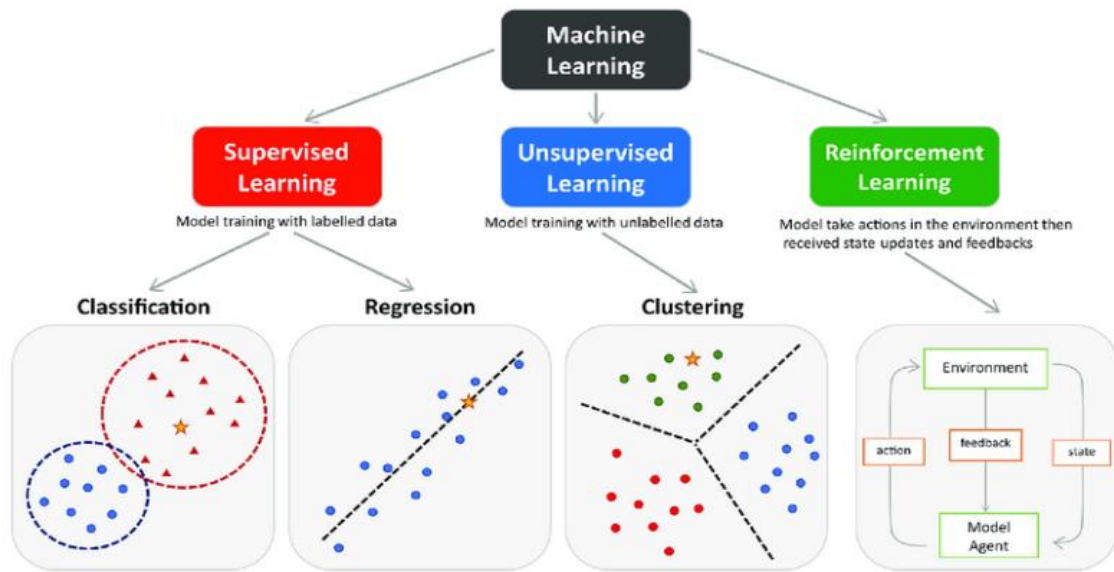


Figure 2 : Subsets of Machine Learning

Each learning paradigm offers distinctive capabilities and finds relevance in diverse contexts, contingent upon the data's characteristics and the objectives of the task under consideration.

2.3.2. Neural Network

Neural networks, essential to AI, merge cognitive science and machine learning principles. These networks simulate biological processes to process information and perform tasks, resembling the brain's neurons. Through interconnected algorithms, they analyze vast datasets to unveil patterns, akin to the brain's pattern recognition. Artificial neurons, or perceptrons (Figure 3), sort data through mathematical operations, while statistical techniques like regression analysis enable applications in forecasting, market analysis, fraud detection, and predictive modeling.

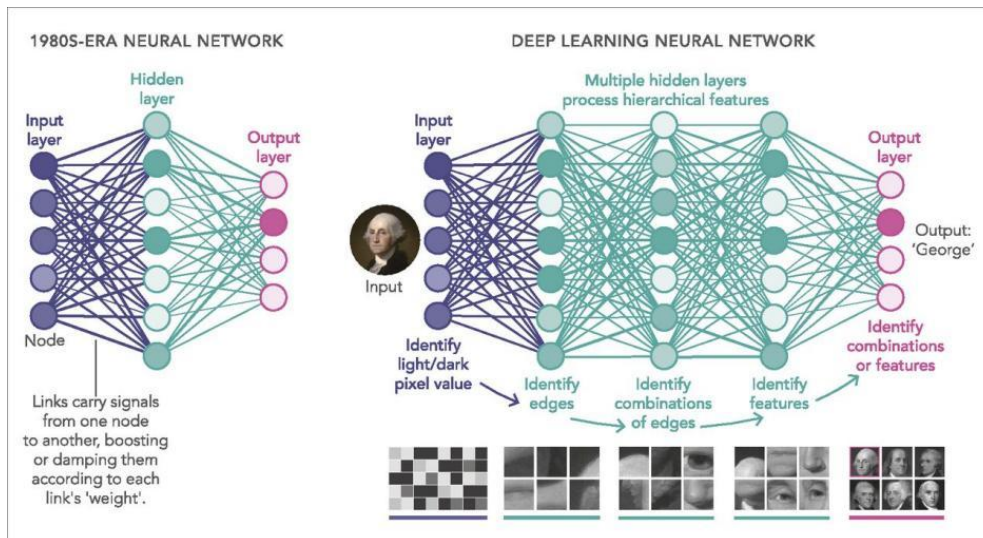


Figure 3 : Architecture of Neural Network

2.3.3. Natural Language Processing

Natural Language Processing (NLP) poses significant challenges for learners, scientists, and engineers, underscored by Turing's test for intelligence assessment. It bridges computer science and AI, facilitating human-computer communication through natural language. By employing computational methods (Figure 4.), NLP enables computers to understand and interpret human language, improving text data processing and insight derivation. Its applications include spam detection, sentiment analysis, and automated summarization, offering benefits like enhanced document accuracy and customer support via chatbots.



Figure 4 :The NLP Process

2.3.4. Computer Vision

Computer vision, a branch of computer science, empowers computers to perceive and interpret visual data from images and videos. Through the use of sensors, artificial intelligence, and deep learning algorithms, computer vision systems analyze patterns

in visual data to recognize and identify objects and people (Figure 5.). This technology finds wide-ranging applications across diverse fields, including text extraction, augmented reality, agriculture, autonomous vehicles, healthcare, and facial recognition.



Figure 5 : Object and People Detection

3. Concept of Reinforcement Learning [7]

In this section, we will thoroughly explore the fundamentals and key concepts of reinforcement learning to elucidate why it's the preferred approach for programming our Robotic Arm.

3.1.1. Definition: What is RL?

RL, or reinforcement learning, is a learning approach where an agent learns through trial-and-error, much like how humans learn new tasks. In RL, the agent interacts with a digital environment, observing its current state and taking actions to maximize a long-term reward. Feedback from the environment informs the agent whether its actions have helped or hindered its progress. Balancing exploration to discover optimal strategies and exploitation of known strategies is crucial for RL agents to achieve their goals effectively (Figure 6.).

3.1.2. Components of RL

a. Agent and Environment

In reinforcement learning, the agent is the main decision-maker, with its behaviors dictated by its policy and learning processes. The environment is the external system

that the RL agent interacts with; it provides the agent with the contextual background against which to make decisions. It responds to the agent's behaviors in a dynamic manner, providing feedback in the form of incentives (rewards) or punishments (penalties) as necessary.

b. Rewards and Penalties

A reward, acting as a scalar feedback signal, communicates the agent's performance level at a given moment, directing its pursuit of a policy that maximizes cumulative rewards, whereas penalties, or negative rewards, are imposed on the RL agent for suboptimal decisions or undesirable behaviors, with the goal of discouraging repetition of such actions in subsequent interactions. In this context, We define two main crucial concepts.

- **Reward Hypothesis:** According to this theory, all objectives in reinforcement learning can be expressed as ways to maximize expected cumulative reward. This indicates that the agent is driven to take actions that will ultimately result in the greatest rewards.
- **Reward Function:** the reward function assigns a positive (reward) or negative (penalty) value to each perceived state (or state-action pair) of the environment. This reward function allows the agent to update its knowledge and enhance its behavior.

c. States and Observations

In RL, a state (s) completely defines the world, whereas an observation (o) just provides a partial picture, perhaps leaving out features. Deep RL often represents states and observations as vectors or matrices. For example, image data such as RGB pixel values or the joint angles and velocities of a robot. The environment is thoroughly observed if the agent observes everything in it. It is partially observed if it observes only portions of the scene.

d. Action Spaces

In RL, actions depend on the environment. The action space is either discrete, with a finite number of moves like in Atari and Go, or continuous, represented as real-valued

vectors, as in robotic control. This difference affects deep RL methods, which need adjustments to switch between discrete and continuous action spaces.

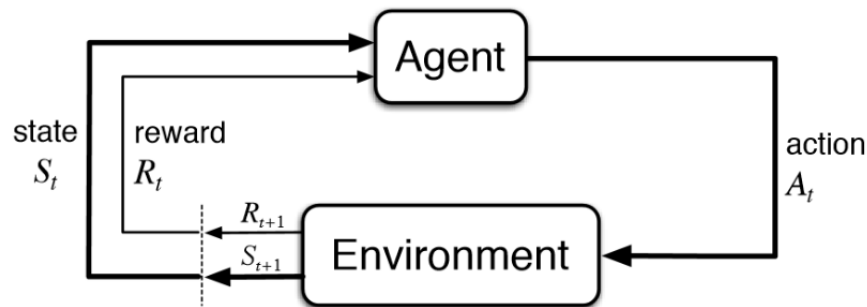


Figure 6 : Flowchart of an RL Algorithm

e. Policies

A policy in RL specifies how an agent should act at a particular time. It plays a crucial role in directing the agent's decision-making process. Formally, a policy implies a mapping between the agent's appropriate actions and the perceived states of the environment. The goal of a policy Based on the environment's context and the agent's present state, the policy dictates the actions the agent takes. It affects the agent's learning process and shapes its behavior.

3.1.3. Theoretical Foundations [8]

a. Markov Decision Processes (MDPs)

Markov Decision Processes (MDPs) are a mathematical framework for describing decision-making in situations where outcomes are both unpredictable and under the control of the decision-maker. MDPs are widely applied in optimization, control theory, and reinforcement learning. In RL, MDPs help us understand how an agent interacts with its environment. At each step, the agent looks at what's happening, picks an action, gets a reward, and moves to a new situation. The aim is to learn a strategy that brings the most rewards over time. Mathematically, we can represent MDP as the tuple (S, A, P, R) , where: S is the set of states, A is the set of actions, P transition probabilities and R rewards. Actually an MDP consists of four key components:

- **States (S)** are a finite set of possible states for the agent. The state collects all relevant information from the history of previous observations, acts, and rewards.
- **Actions (A):** A finite set of acts that the agent may take. The agent's available actions may vary depending on the current situation.
- **Transition Probabilities (P)** (Equation 1.): A function that calculates the probability of arriving at a specific state given a current state and an action. Formally, it is a function $P(s' | s, a)$ that returns the probability of transitioning to state s' if action a is performed in state s .

$$P(s' | s, a) = P(S_{t+1} = s' | A_t = s, A_t = a)$$

Equation 1. Transition Probability

- **Rewards (R):** This gives the immediate feedback the agent receives after taking an action in a certain state. It's represented by a function $R(s, a, s')$, where R is the reward obtained after transitioning from state s to state s' by taking action a .
- **Return** (Equation 2.): The return is calculated as the sum of discounted future rewards, emphasizing the balance between immediate and delayed rewards.

$$G_t = R_{t+1} + R_{t+2} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

Equation 2. Formula of Return

- **Policy** (Equation 3.): A policy guides the agent's action selection at each state independently, with the objective of maximizing the expected return. It is mathematically represented by the formula:

$$\pi(a | s) = P[A_t = a | S_t = s]$$

Equation 3. Policy Formula

This formula calculates the probability of taking action a given the agent's current states, thus defining the distribution of actions over states according to the policy π .

The goal in RL is for the agent to learn a policy, denoted by π , which maps states to actions, such that the cumulative reward over time is maximized. This is often expressed as:

$$\pi^* = \arg \max_{\pi} \sum_{t=0}^{\infty} \gamma^t E[R_t]$$

Equation 4. Optimal policy Equation

Where π^* denotes the optimal policy, γ represents the discount factor (ranging between 0 and 1), R_t signifies the reward at time step t , and E denotes the expected value.

b. Bellman Optimality

The Bellman Optimality Equation is a crucial concept in reinforcement learning. It represents an important relationship that defines the ideal value of a state-action pair in terms of both the immediate reward and the value of the future state. This equation is critical for determining optimal policies and solving reinforcement learning problems effectively.

- State-Value Function: The Bellman Optimality Equation for the state-value function:

$$q_{\pi}(s, a) = R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a \sum_{a' \in S} \pi(a' | s') q_{\pi}(a' | s')$$

Equation 5. State-Value Function

- Action-Value Function: The Bellman Optimality Equation for the action-value function:

$$v_{\pi}(s) = \sum_{a \in A} \pi(a | s) \left(R_s^a + \gamma \sum_{s'} P_{ss'}^a v_{\pi}(s') \right)$$

Equation 6. Action-Value Function:

3.1.4. RL Algorithms [9]

RL algorithms, such as Q-learning or policy gradient methods, are then employed to iteratively update the agent's policy based on the observed rewards and transitions, aiming to converge to the optimal policy.

a. Value-based Algorithms

- **Q-Learning:** Q-Learning, rooted in the Bellman equation, iteratively updates action-value estimates based on observed rewards and future estimates (Figure 7). By iteratively refining these estimates through exploration and exploitation, the algorithm converges towards an optimal policy that maximizes cumulative rewards. This iterative process enables Q-Learning to effectively navigate complex decision spaces, making it a fundamental tool in reinforcement learning for tasks such as robotics and gaming.

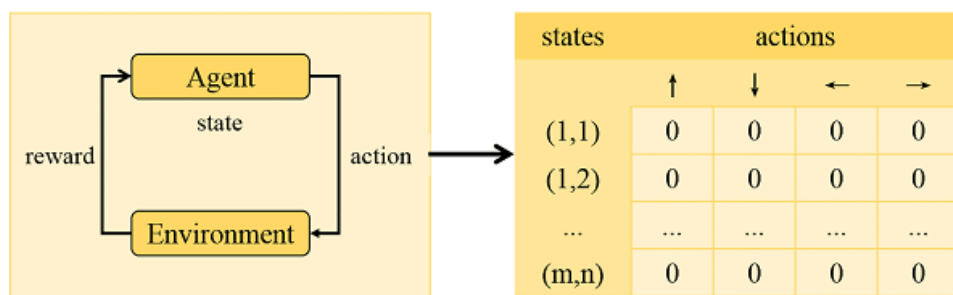


Figure 7 : Learning Architecture

- **Deep Q-Learning Networks (DQN):** Deep Q-Learning, an extension of Q-Learning, incorporates deep neural networks to approximate action-value functions (Figure 8.), enabling it to handle high-dimensional state spaces. By leveraging the power of deep learning, Deep Q-Learning can effectively generalize across states and learn complex strategies, making it suitable for tasks with intricate environments like robotic control and video game playing.

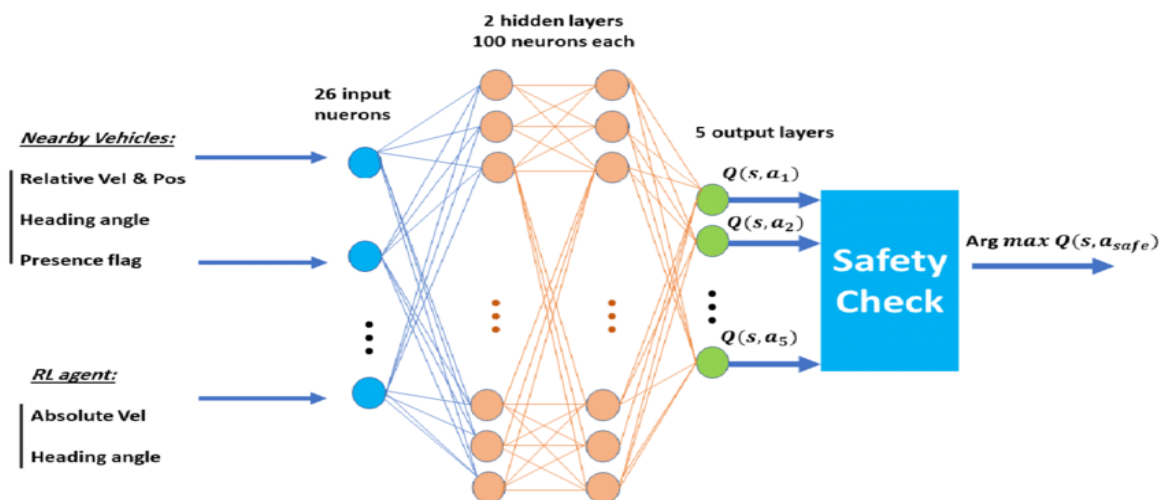


Figure 8 : Deep Q-Learning Architecture

b. Policy-based Algorithms

- **Policy Gradient Methods:** Policy Gradient Methods optimize the policy directly, learning a parameterized policy that selects actions without needing a value function (Figure 9). They estimate the gradient of expected return and update policy parameters in the direction that increases expected return. This approach enables handling continuous action spaces and stochastic policies effectively.

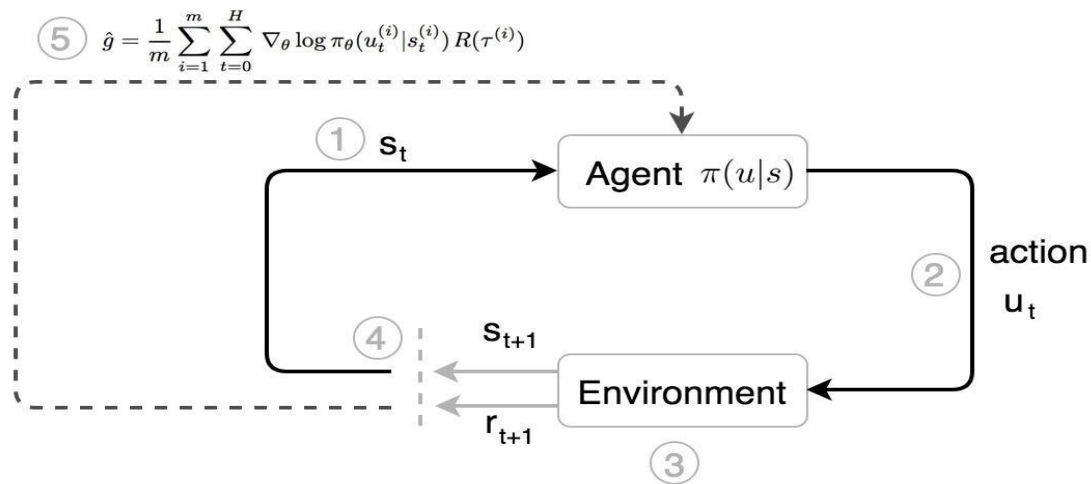


Figure 9 : Policy Gradient Methods Concept

- Actor-Critic Methods: They combine both policy-based and value-based approaches, where the actor network learns the policy, and the critic network evaluates the actions taken by the actor to update the policy (Figure 10). This method enhances learning stability and efficiency by leveraging both value estimation and policy optimization simultaneously.

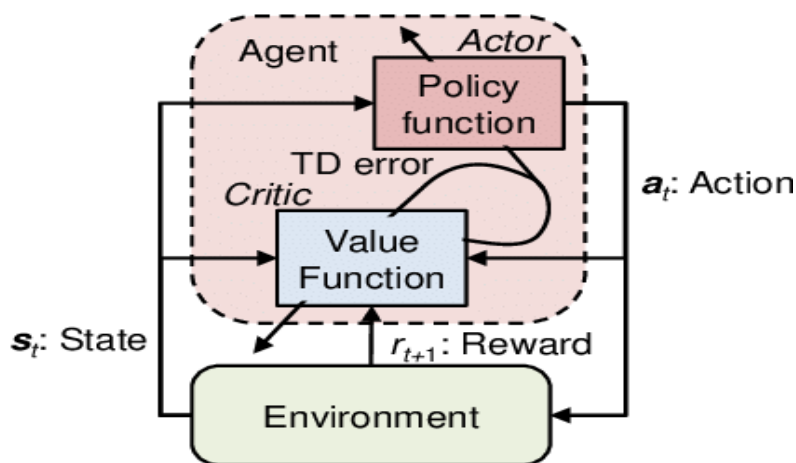


Figure 10 : Actor-Critic Method's Concept

Conclusion

This chapter has laid down the theoretical foundations of AI and RL, providing an essential conceptual framework for the rest of the report. We have seen how these concepts are fundamental for the development of autonomous systems capable of learning and adapting their behavior in dynamic environments.

Chapter2: Robotic Arms - Functionalities and Applications

Introduction

This chapter delves into the theory and techniques of robotic arm control, spanning both traditional and advanced methods. Robotic arms play a vital role in automation, driving innovation and efficiency across various industries, from manufacturing to healthcare. Their adaptability and functionality are instrumental in advancing technological capabilities

1. Historical Development [3]

Robotic arms, originating from the Unimate in the 1950s, revolutionized industrial tasks on General Motors' assembly line by handling hazardous jobs. Advancements in microprocessors and computer technology in the 1970s expanded their precision and use in space exploration, such as the Canadarm. Integration of robotics with reinforcement learning (RL) enabled arms to learn from their environment, extending their applications to healthcare and research. Subsequent decades brought enhancements with computer programming and electronic servo motors, allowing for more complex operations (Figure 11). Today, RL integration drives autonomy and adaptability, enabling robotic arms to continually improve efficiency and functionality.

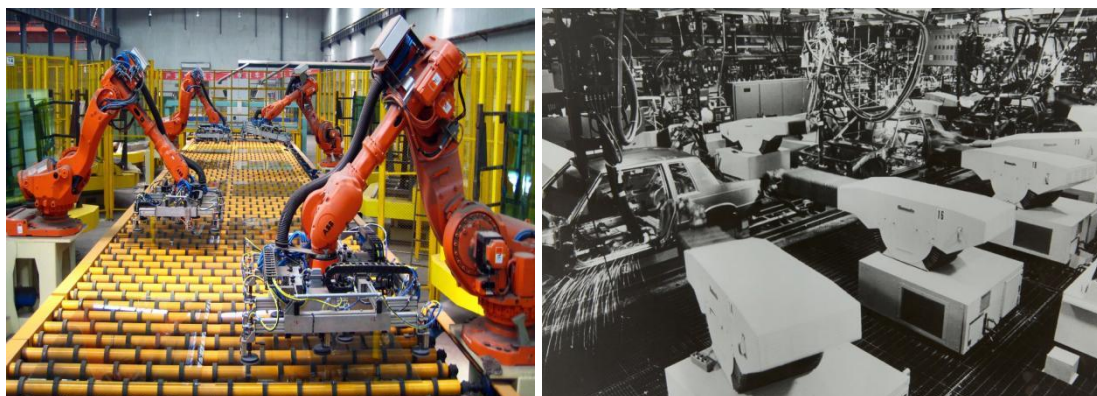


Figure 11 : Robotic Arms Through History

Basic Components and Mechanisms [4]

Robotic arms may perform a variety of functions thanks to hardware and software integration.

1.1. Hardware

In the hardware domain, components like the arm structure and actuators shape the robotic arm's manipulation capabilities. The arm's structure must be sturdy yet flexible. A robotic arm is a complex piece of machinery, typically composed of several key components (Figure 12):

- **Base:** The base is the foundation of the robotic arm. It provides stability and is typically fixed to a solid surface. The base also often includes a motor that allows the arm to rotate.
- **Joints:** Joints connect the various parts of the robotic arm and allow for movement. The number and type of joints determine the degrees of freedom of the arm, i.e., the directions in which it can move.
- **The end-effector:** is the device at the end of a robotic arm that interacts with its surroundings. It could be a gripper, a welding torch, a camera, or any other equipment, depending on the application.

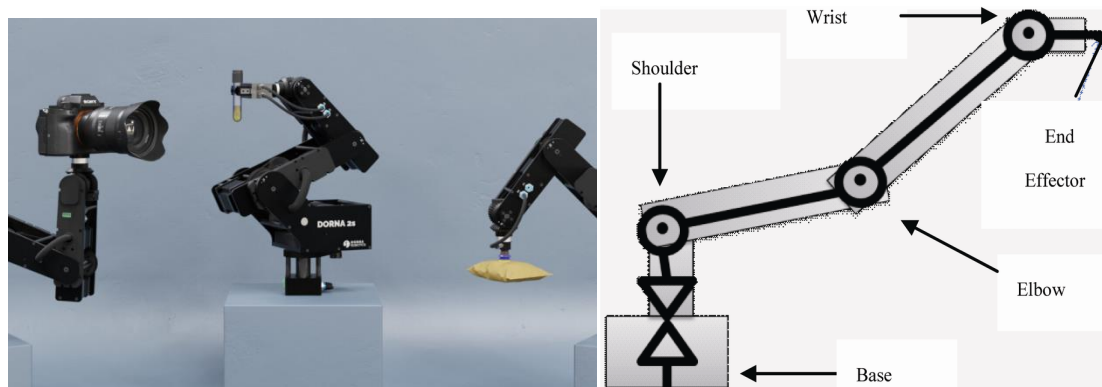


Figure 12 : Different Components of Robotic Arm

Designing a robotic arm requires careful consideration of several factors, we mention some of them in the Table 1 below:

Table 1. Requirements of Designing a Robotic Arm

Factor	Description
Structural Integrity	The arm must withstand operational forces without breaking or deforming.
Weight Optimization	Optimizing weight is crucial for speed, power consumption, and load handling capabilities
Material Selection	Consider factors like strength, weight, cost, and compatibility for
Kinematics	Determines arm movements through joint angles and segment lengths.
Actuation Systems	These systems, such as electric motors or hydraulics, move the arm

1.2. Control Systems and Software

Robotic arms utilize various control systems to manage their movements and interactions with the environment. These systems can be broadly categorized into two types:

- **Open-loop Control** (Figure 13): Predefined commands are sent to the actuators without considering feedback from sensors. While simple, this approach lacks adaptability to changing conditions.

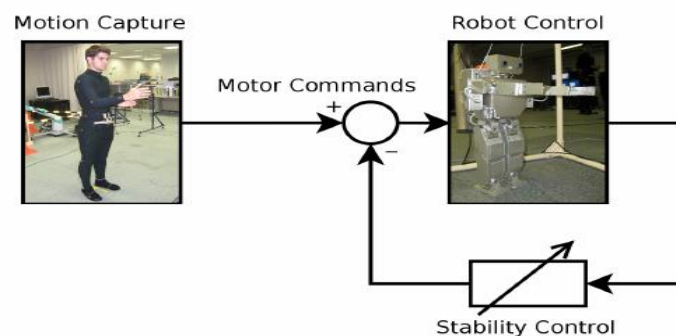


Figure 13 : Open-loop Control

- **Closed-loop Control** (Figure 14): Feedback from sensors is used to adjust the arm's actions in real-time, ensuring precise positioning, force application, and interaction with objects. This approach enhances accuracy and robustness.

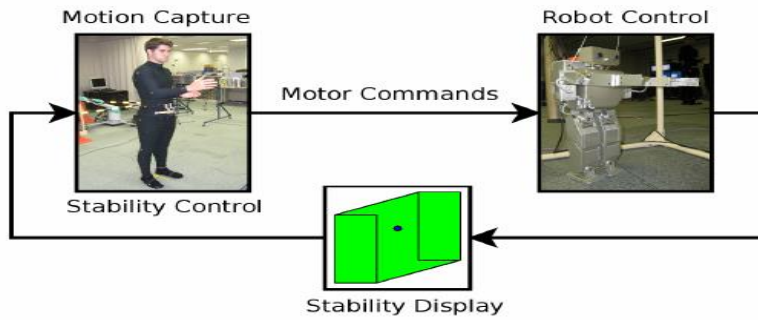


Figure 14 : Closed-loop Control

- **Software Integration:** Advanced software algorithms are integrated into robotic arms to enhance their capabilities and adaptability. One notable technique is Q-learning, to enable the arm to learn optimal actions through trial and error:

1.3. Kinematic Study of Motors [5]

1.3.1. kinematics of robotic arms and task execution

a. Definition

Kinematics is a branch of mechanics, focuses on understanding motion without delving into the forces causing it. In robotic arms, kinematics explores how dimensions and connections within the arm relate to the position, speed, and acceleration of its components.

b. Types of Kinematics [6]

Robotic arms utilize two main types of kinematics (Figure 15.):

- **Forward Kinematics:** This computes the position of the arm's end effector based on the angles of its joints.
- **Inverse Kinematics:** This reverses the process, determining the necessary joint angles to achieve a desired end effector position.

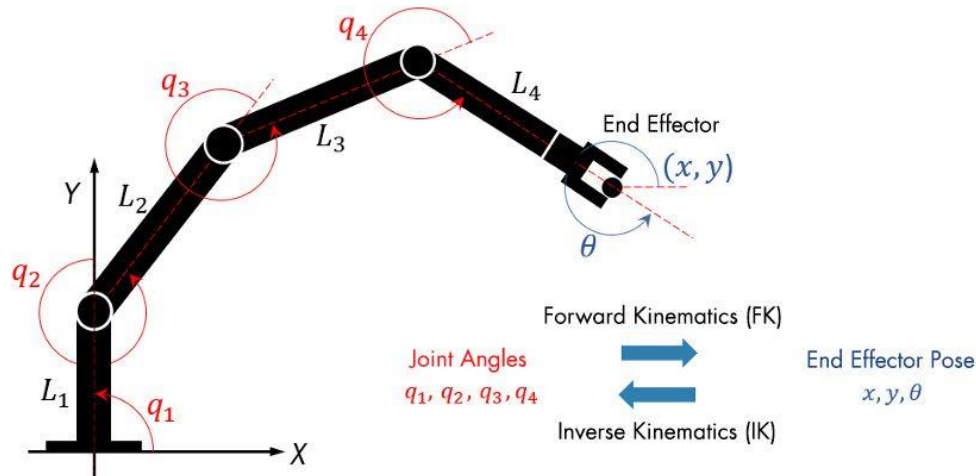


Figure 15 : Types of Kinematics

c. Task Execution

Robotic arms leverage their kinematics, motion planning algorithms, and sensory input to execute various tasks. Equipped with advanced vision systems, they can swiftly recognize, grip, and transport objects as needed. Tasks may include reaching, grasping, lifting, rotating, or manipulating objects, with the control system continuously adjusting based on sensor feedback to ensure accuracy and safety.

1.3.2. types of motors used and their kinematic models.

In robotic arms, various types of motors are utilized, each with its corresponding kinematic model. DC motors, stepper motors, servo motors, hydraulic actuators, and pneumatic actuators commonly power robotic arms, with their movements typically described using joint-space models.

2. Applications Across Industries

Robotic arms find extensive applications across various industries, revolutionizing manufacturing processes, healthcare, agriculture, aerospace, and beyond. In manufacturing, robotic arms streamline assembly lines, enhancing efficiency and precision, as seen in automotive production. In healthcare, they facilitate minimally invasive surgeries, such as robotic-assisted surgeries for intricate procedures like prostatectomies. In agriculture, robotic arms aid in tasks like crop harvesting and

sorting, optimizing yield and reducing labor costs. In aerospace, they contribute to spacecraft assembly, where precision and reliability are paramount. These examples underscore the versatility and impact of robotic arms across industries, driving innovation and productivity.



Figure 16 : Applications Across Industries

3. Challenges and Future Directions

Among the obstacles that a robotic arm faces, we can list:

- **Robotic arms have limited dexterity:** when it comes to sensitive activities like fine motor skills. To solve this, better end-effectors and smarter control systems are required for accurate movements.
- **Decision Complexity:** They struggle to make decisions in cluttered environments. Making them smarter through AI and learning can aid in their understanding and adaptation to their surroundings.
- **Human Interaction:** Robotic arms must be able to operate closely with humans while remaining safe. Future models could be created as cobots, which can perceive and communicate better, resulting in smooth teamwork.

Conclusion:

In This chapter, we have explored various aspects of robotic arm control, from kinematics and dynamics to the utilization of both traditional and advanced control techniques. It has laid the necessary groundwork for comprehending the challenges and opportunities linked to designing efficient control systems for robotic arms.

Chapter 3: Design and Implementation of a Reinforcement Learning-based Robotic Arm

Introduction:

In This section we provide an overview of the chapter's objectives, focusing on the practical aspects of designing and implementing a robotic arm using reinforcement learning (RL) techniques. It outlines the significance of utilizing RL in robotic systems and sets the stage for the subsequent sections.

1. Mecanical conceptualization:

1.1. CATIA

CATIA (Figure 17), developed by Dassault Systèmes, is a comprehensive CAD software suite used across various industries for 3D design, engineering, and simulation. It offers robust tools for creating intricate 3D models with high geometric precision.



Figure 17 : CATIA Logo

The software also provides digital prototyping and simulation capabilities, allowing engineers to virtually analyze a product in its operating environment (Figure 18). Integrated with Product Lifecycle Management (PLM) systems, CATIA facilitates efficient data management and collaboration.

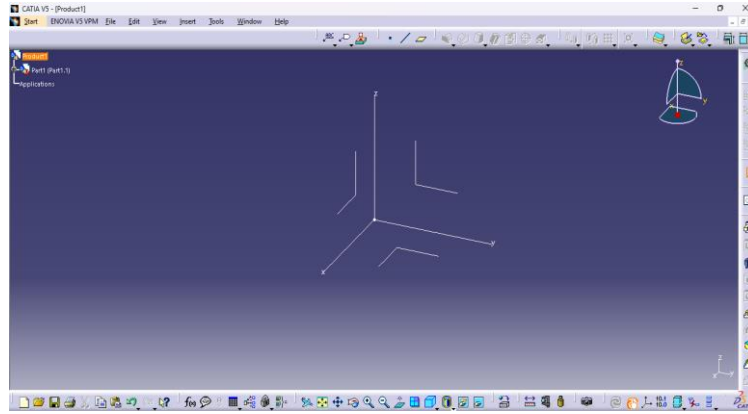


Figure 18 : The CATIA Interface

1.2. Robotic Arm Design

1.2.1. Inspiration:

Our design philosophy was profoundly shaped by the concept of servo motors, renowned for their precise control over angular position. Additionally, we drew inspiration from the mechanical conception of servo motors, recognizing their compact yet powerful design as a model for efficiency and reliability in our robotic arm. Drawing inspiration from their functionality, we aimed to create a robotic arm design that harnesses this precision while minimizing complexity. Utilizing CATIA, we creatively modeled each component, ensuring seamless integration with servo motors and prioritizing ease of assembly. By embracing the simplicity inherent in servo motor technology, we optimized the design process to achieve both structural integrity and functional efficiency in our robotic arm.

.

1.2.2. Different Parts of The Robotic Arm

In this section, we will delineate the various components of the robotic arm that we conceptualized (see appendix), accompanied by images of each part. Additionally, we will specify the length and overall dimensions of the arm, with detailed information available in the appendix. In the figures below: Figure 19 and Figure 20, we delineate the distinct components of our robotic arm and illustrate the assembly in the Figure 21 provided below.

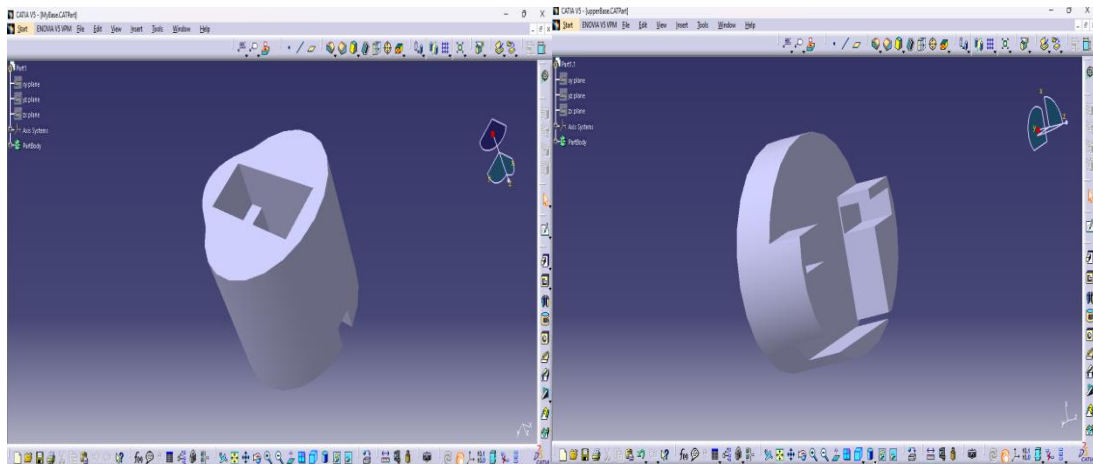


Figure 19 : Base and Upper Base

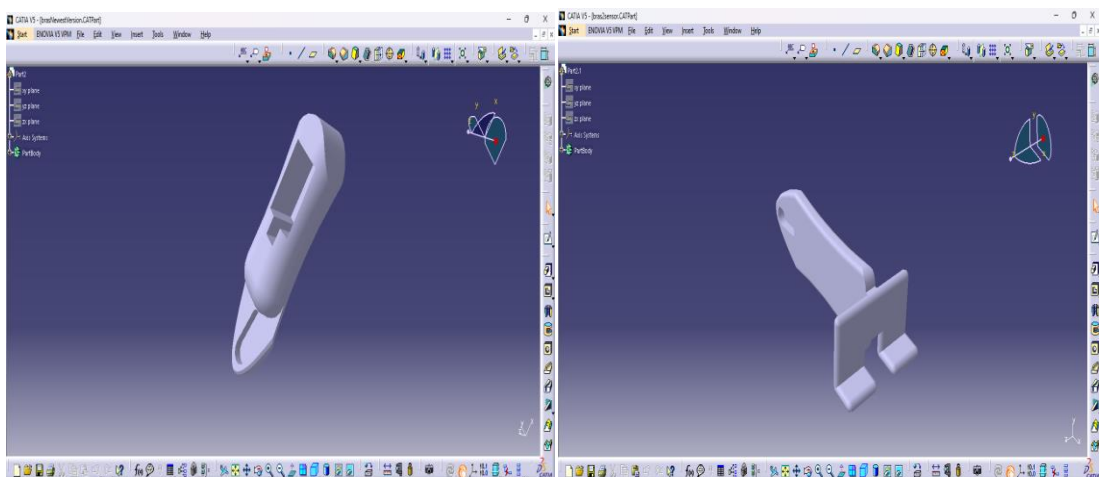


Figure 20 : Shoulder and Elbow

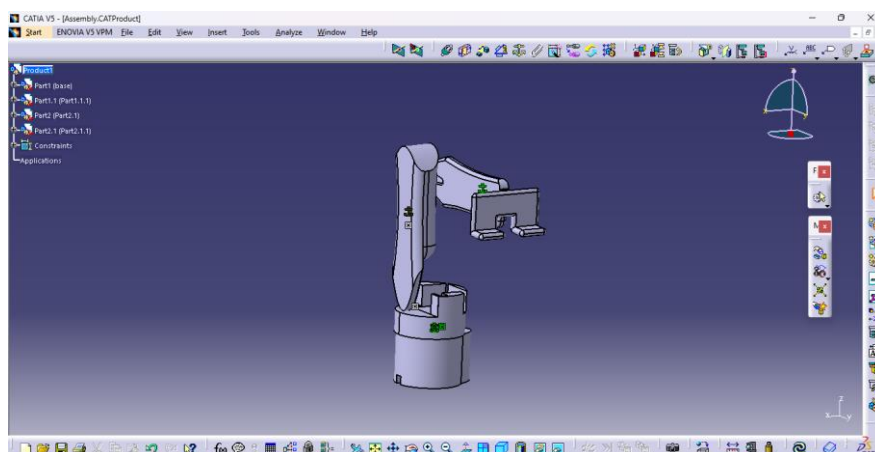


Figure 21 : Assembly of The Robotic Arm

1.2.3. The Slicing and Printing Process

A. Presentation of the 3D printer

The Raise3D (Figure 22) printer is a versatile and high-performance additive manufacturing machine, known for its precision, reliability, and large print volume. It is manufactured by Raise3D, a leading company in the 3D printing industry. Its key features include:

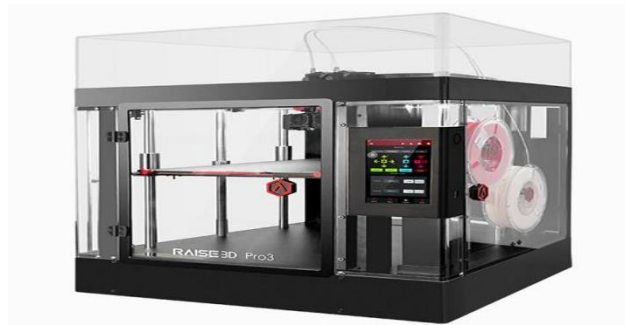


Figure 22 : Raise3D Printer

Large Print Volume: Raise3D printers are available in various sizes with generous build volumes, allowing users to produce large and complex parts in a single print.

High Precision: These printers can achieve high levels of print accuracy and detail, making them suitable for a wide range of applications, including prototyping, manufacturing, and engineering.

Dual Extrusion: Many models of Raise3D printers come equipped with dual extruders, enabling multi-material and multi-color printing.

Robust Construction: Raise3D printers are built with robust and durable components, ensuring stability and reliability during the printing process.

User-Friendly Interface: These printers feature intuitive touchscreen interfaces and user-friendly software, making them accessible to both beginners and experienced users.

Compatibility with Various Materials: Raise3D printers support a wide range of filament materials, including PLA, ABS, PETG, TPU, and more.

Overall, Raise3D 3D printers are valued for their combination of advanced features, reliability, and user-friendly design, making them popular choices in the 3D printing community.

B- The 3D modeling software

In our project, we utilized slicing software such as IdeaMaker (Figure 23), a powerful tool essential for preparing 3D models for printing. Slicing involves breaking down a 3D model into layers and generating the necessary toolpaths for the 3D printer to follow. This process is crucial for ensuring optimal print quality and structural integrity. The software allows us to customize various printing parameters such as layer height, infill density, and print speed to achieve the desired balance between print quality and efficiency.

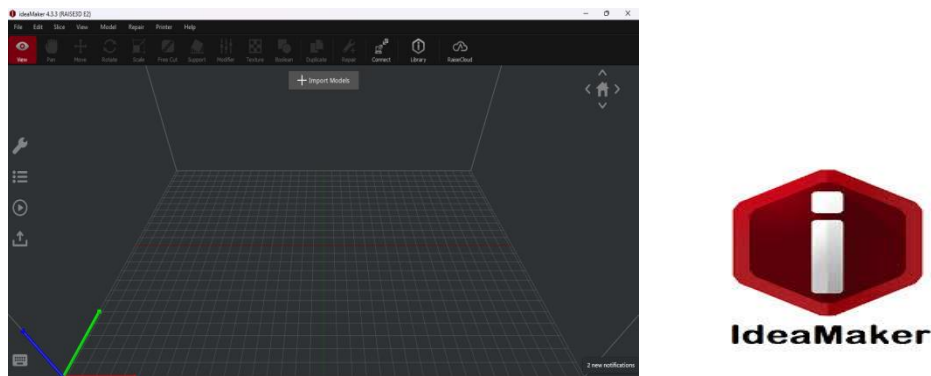


Figure 23 : IdeaMaker Interface and Logo

C- printing methods and choice of materials

Polylactic Acid (PLA): PLA is a biodegradable thermoplastic filament derived from renewable resources such as corn starch or sugarcane (Figure 24). It is one of the most used materials in 3D printing due to its ease of use and environmentally friendly nature. It exhibits low warping and good adhesion to the print bed, ensuring reliable and consistent prints.



Figure 24 : Polylactic Acid (PLA)

Environmental Friendliness: PLA is biodegradable and sourced from renewable resources, making it an environmentally friendly choice for 3D printing. It does not emit harmful fumes during printing, unlike some other thermoplastics, making it safe to use in home or office environments.

Wide Range of Colors: PLA filament is available in a wide variety of colors, allowing for creativity and customization in 3D printing projects. It can also be easily painted or post-processed to achieve the desired aesthetic.

Strength and Rigidity: Although PLA is not as strong or heat resistant as some other 3D printing materials like ABS or PETG, it still offers good strength and rigidity for many applications.

We selected the Standard PLA 1.75mm filament due to its compatibility with our 3D printer and its affordability. This choice aligns with our goal of making practical decisions accessible to everyone involved in the project.

D- Slicing Process

-Transforming CAD Designs into Printable Blueprints: The Role of STL Files

STL, which stands for stereolithography, is a standard file format widely used in 3D printing. It represents the surface geometry of a 3D object using a series of interconnected triangles. This format is commonly accepted by most 3D printers and slicing software. STL files essentially provide a blueprint for the 3D printer, guiding it on how to construct each layer of the object. They contain information about the shape, size, and position of every surface, enabling precise reproduction of the original 3D model.

- Start Slicing

Initiating the slicing process involves strategically positioning the individual components on the 3D printer plate (Figure 25), which measures 150mm x 150mm. It's essential to ensure that each piece fits within the designated area to prevent any parts from extending beyond the plate boundaries. This meticulous placement guarantees optimal printing efficiency and accuracy.

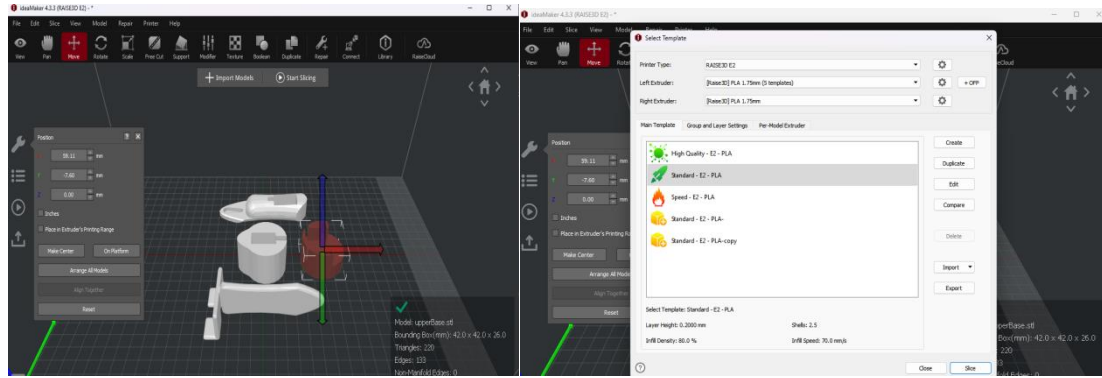


Figure 25 : Configuration of the Slicing Type

We display the "Estimated Print Result" (Figure 26), indicating a total print time of 16 hours, 11 minutes, and 57 seconds for the entire assembly. This estimation serves as a valuable reference for planning and scheduling the printing process, providing insights into the expected duration of the printing operation.

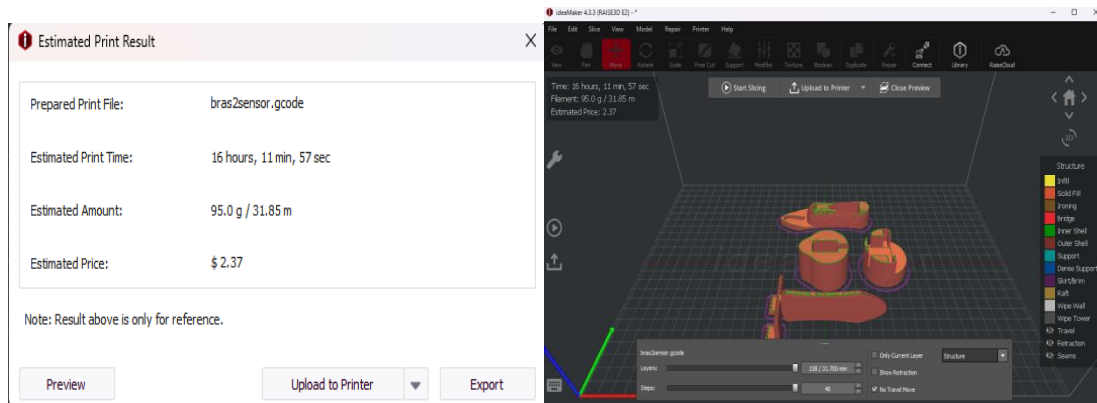


Figure 26 : Results of Slicing

It's possible to reduce the printing time of the pieces by increasing the slicing speed, but this could impact the final accuracy of our robot.

2. Workspace Environment

2.1. Hardware

2.1.1. Raspberry Pi Board: Features and functionalities [10]

The Raspberry Pi 3B+ (Figure 27) is a single-board computer developed by the Raspberry Pi Foundation. It features a Broadcom BCM2837B0 chipset with a 64-bit

quad-core ARM Cortex-A53 processor running at 1.4 GHz. The board includes 1GB LPDDR2 SDRAM, wireless LAN, Bluetooth 4.2, Gigabit Ethernet, HDMI output, four USB 2.0 ports, and a MicroSD card slot for storage.

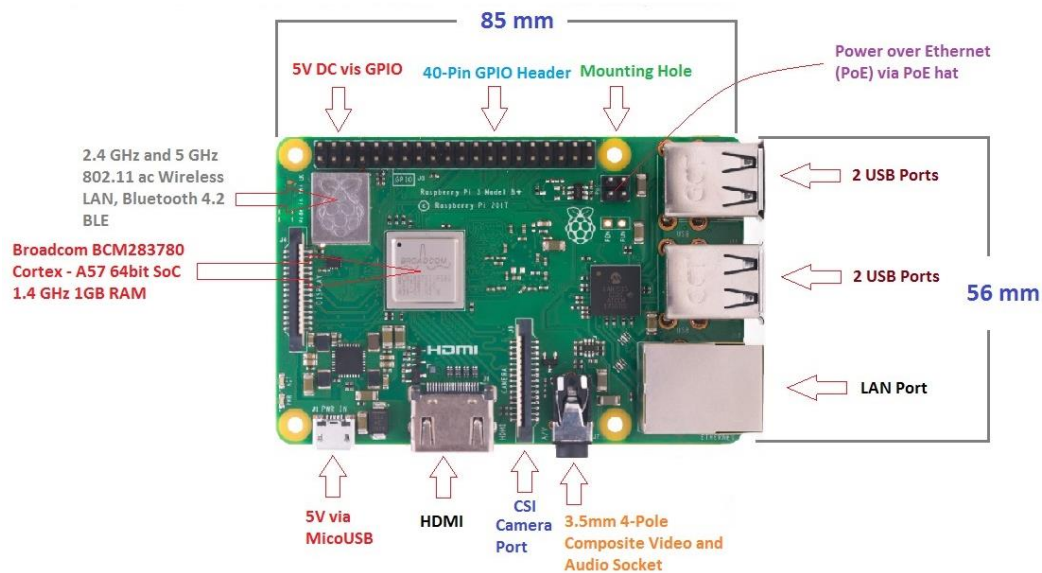


Figure 27 : Raspberry Pi Board

2.1.2. Operating System: Configuration and selection

• Linux-based Operating Systems

Raspbian: It is the reference operating system for Raspberry Pi (Figure 28). It is based on Linux Debian (raspberrypi + debian = raspbian) and is regularly updated. In 2021, Raspbian changed its name to Raspberry Pi OS. The old name was easier to remember, but the new version is much easier to install and use. It is specially optimized for Raspberry Pi, providing a versatile distribution that allows us to easily familiarize ourselves with the hardware.



Figure 28 : Raspbian OS logo

2.1.3. SD memory card class 10, 32GB

This is where the operating system of our Raspberry Pi and all our software will be stored. With the need to store libraries and the complexity of the algorithm, we have chosen 32GB (Figure29), to ensure sufficient storage space.



Figure 29 : SD memory card

2.1.4. Servo Motors Micro Servo 9g SG90 Mini Servo 180 Degree Rotation Plastic Gear

We opt for this type of servos (Figure 30) because stands out as a popular choice in the market due to its widespread availability, affordability, and user-friendly design, making it a common option for various projects. A servo motor is a type of rotary actuator that allows for precise control of angular position . Specifications are provided in the table below:

Table 2. Specifications Of Servo Motors

Feature	Description
Precision Control	Servo motors offer precise control over angular position, ideal for applications requiring accurate positioning.
Feedback Mechanism	Equipped with feedback mechanisms like encoders, providing real-time position information to the control system.
High Torque	Provides high output torque relative to their size, suitable for applications requiring both precision and power.
Variable Speed	Capable of operating at variable speeds, enabling smooth and controlled motion.
Compact Size	Compact design makes them suitable for applications with limited space.
Compatibility	Interfaces with various control systems and microcontrollers, facilitating integration into different projects.



Figure 30 : Servo Motor

2.1.5. Ultrasonic Sensor Sensor Module HC-SR04

We opt for ultrasonic sensor module HC-SR04 (Figure 31) because it is the most commonly used modules due to its widespread availability, ease of use, and cost-effectiveness. An ultrasonic sensor is an instrument that measures the distance to an object using ultrasonic sound waves. An ultrasonic sensor uses a transducer to send and receive ultrasonic pulses that relay back information about an object's proximity.



Figure 31 : Ultrasonic sensor

2.1.6. Raspberry Pi Wiring for Robotic Arm Control

Wiring the Raspberry Pi to 3 servo motors and an ultrasonic sensor using an internal power supply provided by our nano-computer usinh Fritzing (Figure32).

- **Servo Motors Connection:**

- Connect the power wires (+) of the servo motors to a 5V power pin of your Raspberry Pi.
- Attach the ground wires (-) of the servo motors to a ground (GND) pin of your Raspberry Pi.
- Utilize available GPIO pins of your Raspberry Pi to control the signals of the servo motors. Ensure to choose pins compatible with PWM (Pulse Width Modulation) for servo motor control.

- **Ultrasonic Sensor Connection:**

- Power the ultrasonic sensor using a 5V power source provided by the Raspberry Pi, also connecting it to the common ground with the servo motors.

- Wire the signal pins of the ultrasonic sensor (trig and echo) to available GPIO pins of the Raspberry Pi for signal transmission and reception.

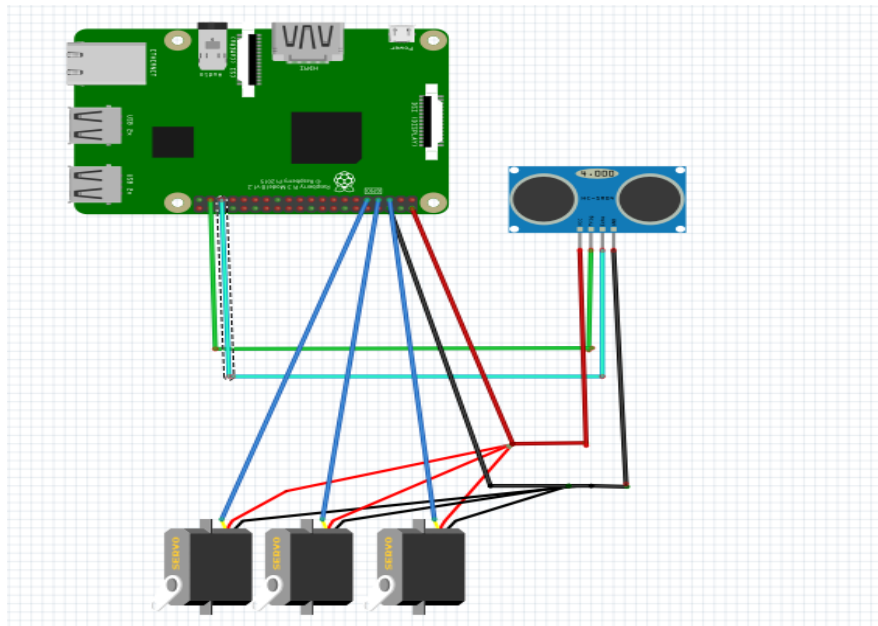


Figure 32 : Servo Motors Connection

2.2. Software Environment Setup:

2.2.1. Software Environment Selection

A- Remote Desktop Access Systems: PuTTY and VNC

Remote desktop access systems, such as PuTTY (Figure33) and VNC, facilitate remote interaction with computers and servers. PuTTY offers text-based access for managing systems via SSH, Telnet, and serial connections, while VNC (Figure 34) provides graphical access to remote desktop environments, enabling users to control them from afar.

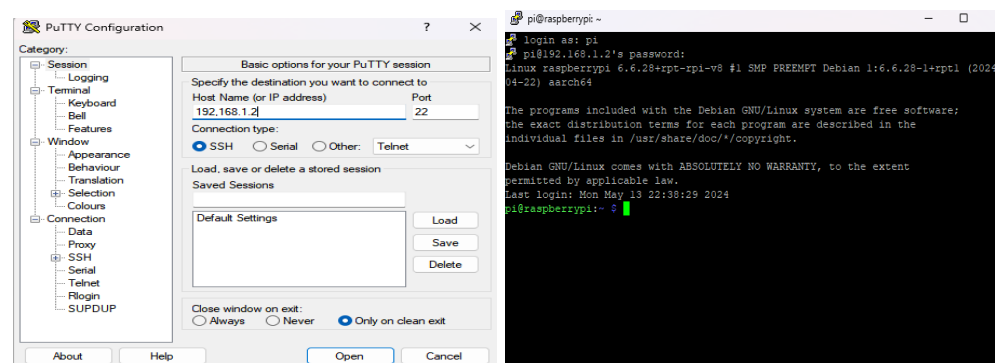


Figure 33 : Putty Configurations

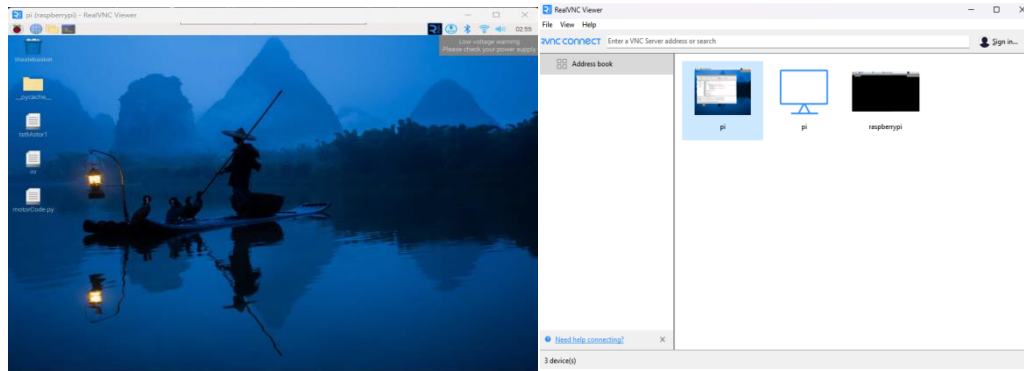


Figure 34 : VNC Graphical Interface

B- Python [6]

Python is a high-level interpreted programming language known for its simplicity and readability. It offers a vast ecosystem of libraries and frameworks, making it ideal for a wide range of applications. Its main advantages include ease of use, rapid prototyping capabilities, and seamless integration with deep learning frameworks like TensorFlow and PyTorch. (Figure 35).

In the table below we provide a comparative description of the two frameworks.

Table 3. Comparison between Tensorflow and PyTorch

Framework	Description
TensorFlow	TensorFlow is an open-source deep learning Framework developed by Google that offers a comprehensive platform for building and training machine learning models, particularly deep neural networks ,with a focus on scalability and flexibility
PyTorch	PyTorch is a Python-based open-source deep learning framework that provides a flexible and intuitive interface for building and training neural networks



Figure 35 : TensorFlow Python PyTorch

C- Visual Studio Code (VSCode)

In the context of our project, the Python execution environment in Visual Studio Code (VS Code) (Figure 36) offers several advantages for implementing reinforcement learning (RL) techniques. Its versatile IDE provides a conducive environment for coding, debugging, and testing RL algorithms, enabling us to iterate quickly and refine our models effectively.



Figure 36 : Visual Studio Code

D- Fritzing: Wiring simulations

Fritzing is an open-source software used for designing electronic circuits, creating printed circuit board (PCB) layouts (Figure 37), and making electronic assembly drawings. Fritzing allows users to draw electrical schematics using a library of predefined electronic components, create PCB layouts from these schematics, and visualize electronic assemblies graphically.



Figure 37 : Fritzing

3. Implementation of Reinforcement Learning:

3.1. RL Algorithm Selection: DQN

We chose the Deep Q-Learning algorithm for a variety of reasons. We list some of the most privileged features that the DQN algorithm offers:

- **Efficiency in Complex Environments:** DQN excels at problems involving high-dimensional state and action spaces, which are ubiquitous in robotics applications. Its capacity to approximate the value function with deep neural networks allows it to deal with complicated contexts successfully.

- **Real-Time Adaptability:** DQN supports online learning, allowing it to modify and refine its policy in real time as it interacts with its surroundings. This allows the robotic arm to continuously adjust its control strategy based on real-time feedback obtained during operation, improving its adaptability to dynamic circumstances.
- **Generalization Across activities:** DQN has showed the ability to apply learnt policies to similar activities and contexts. This means that once taught, the DQN-controlled robotic arm can use its acquired knowledge to efficiently handle new circumstances without requiring lengthy retraining, saving time and costs.

3.2. Results and Evaluation:

3.2.1. Project of 2D Simulation Environement of DQN RL-based robotic arm

In this project, we focused on creating a simplified 2D simulation environment, using VSCode, to facilitate learning for less complex agents, serving as a precursor to more advanced robotic arm simulations. By simplifying the scenario, we aimed to provide a foundational understanding of reinforcement learning principles in a controlled context. This approach streamlined the learning process for both researchers and agents, paving the way for smoother transitions to more challenging tasks. Our simulation aimed to bridge theoretical concepts with practical implementation, enabling a gradual progression towards handling complex scenarios in robotic arm control.

Our project, showcases the Deep Q-Learning (DQN) algorithm applied to solve target-reaching tasks in a 2D robot arm scenario (Figure 38).

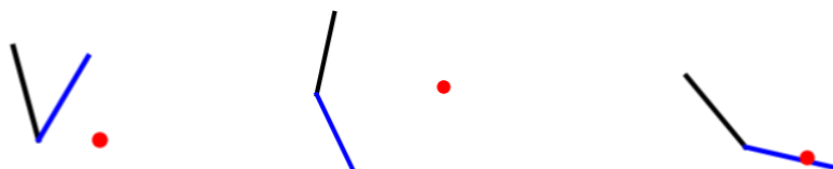


Figure 38 : Agent

We provide below the implementation outcomes (Figure 31):



Figure 39 : performance of The model

Analysis of The Figure 31:

- **Reward vs Episode Graph:** The graph shows the learning progress of a DQN algorithm controlling a 2D robotic arm. The x-axis represents the number of episodes, and the y-axis represents the reward values.
- **Total Reward Trend:** The blue line indicates the total reward per episode. Despite fluctuations, there's an overall upward trend, suggesting that the DQN is learning and improving its control strategy over time.
- **Average Reward Trend:** The orange line shows the average reward, which smooths out the fluctuations of the total reward. It starts lower but gradually improves, indicating consistent learning progress.
- **Learning Insights:** The increasing trend in both total and average rewards implies that the DQN model is effectively learning the task. The fluctuations in the total reward could be due to exploration or the complexity of the task.

3.2.2. Project of Our Real Robotic Arm

In extending our project from a 2D to a 3D robotic arm (Figure 40), we aimed to enhance the complexity and realism of our simulation. By transitioning to a 3D space, we introduced additional challenges such as increased degrees of freedom and spatial

navigation. This expansion allowed for a more comprehensive exploration of reinforcement learning principles in robotics, providing valuable insights into the adaptation of algorithms to diverse environments. Our evaluation focused on measuring the performance of the 3D robotic arm in completing tasks, assessing its ability to generalize learned behaviors, and identifying areas for further improvement in real-world applications.

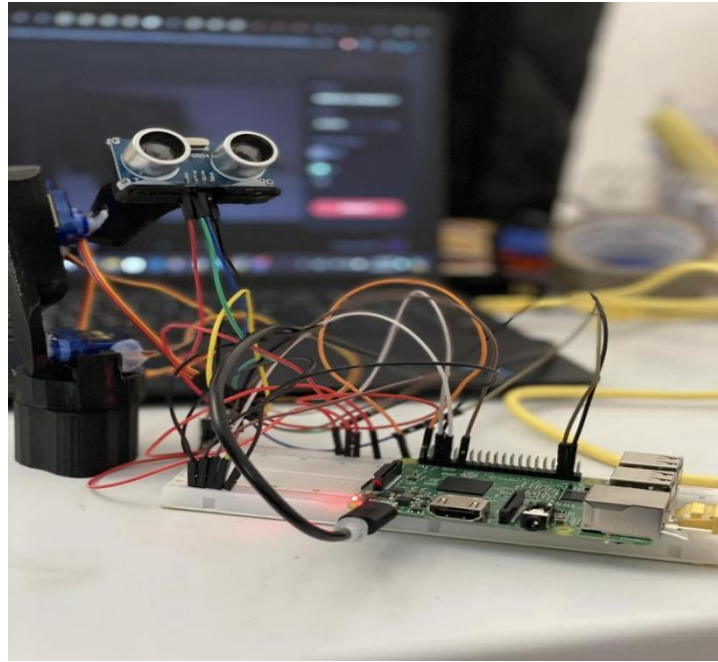


Figure 40 : Our Robotic Arm

Conclusion:

Throughout this chapter, we have explored the design and implementation of a reinforcement learning-based robotic arm. We have demonstrated the practical steps involved in creating the physical components, establishing the necessary hardware and software environment, and implementing the Deep Q-Learning algorithm. Our findings highlight the potential of reinforcement learning in enhancing the autonomy and performance of robotic systems, paving the way for future advancements in robotics research and development.

GENERAL CONCLUSION

In this project, we've explored how combining artificial intelligence (AI) with robotics can make robotic arms smarter and more capable. We started by looking at the basics of AI, focusing on reinforcement learning (RL), which teaches machines to learn from experience. Then, we delved into robotic arms, understanding their parts, how they've evolved, and what they can do in different industries.

Next, we got hands-on, designing and programming a robotic arm using RL techniques. This involved using software to design the arm and setting up the right tools for training it. We chose a specific RL algorithm called Deep Q-Learning because it's good at handling complex tasks.

After that, we tested our RL-based robotic arm in different scenarios, both in simulations and in real life. We found that it performed well, showing promise for future applications in industries like manufacturing and healthcare.

Looking ahead, this project highlights the exciting potential of combining AI and robotics. By teaching machines to learn and adapt, we can create smarter, more efficient tools that have a big impact on how we work and live. With further research and development, RL-based robotic arms could revolutionize various industries, leading to safer, more productive workplaces and new opportunities for innovation.

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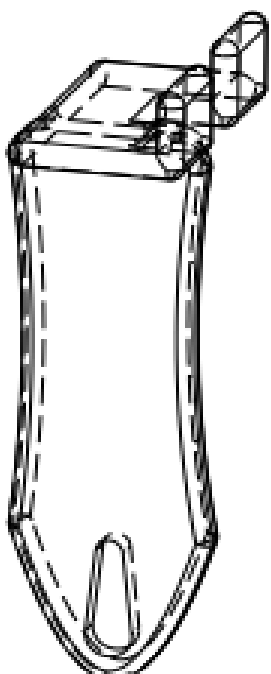
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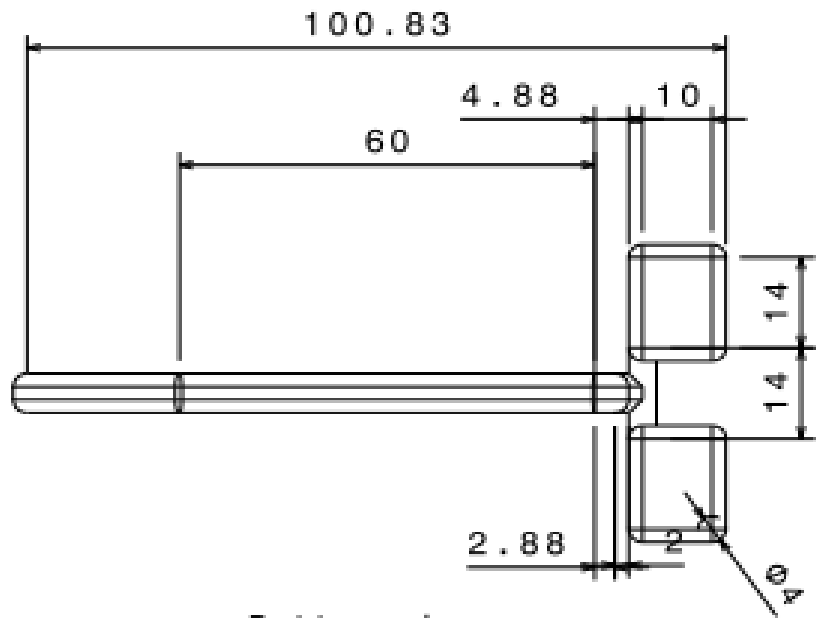
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APPENDIX

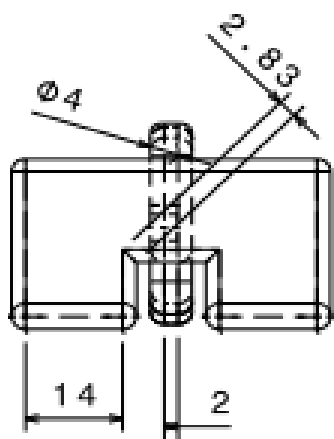
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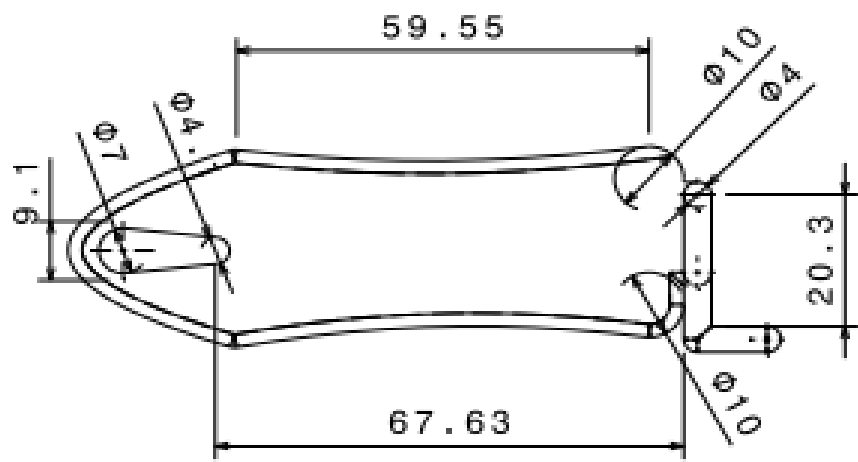
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Right view
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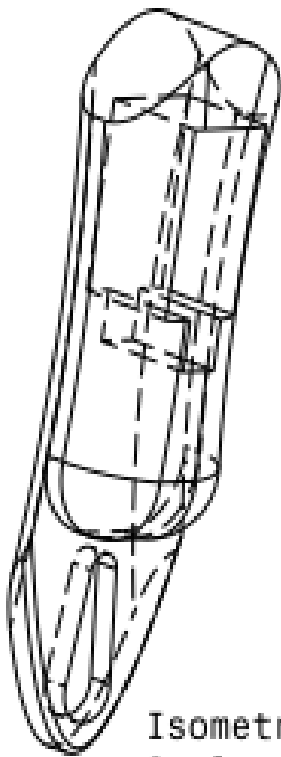


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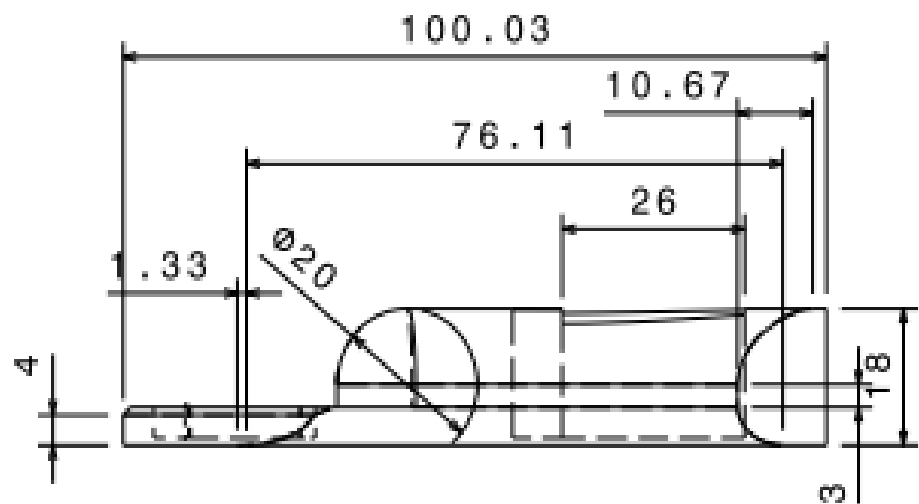
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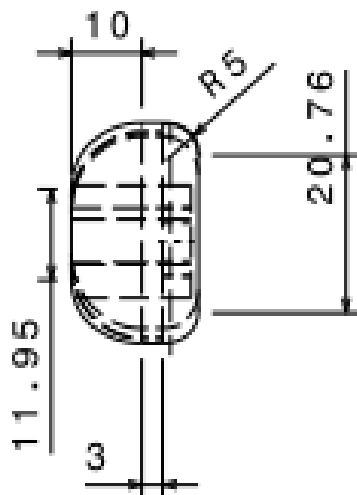
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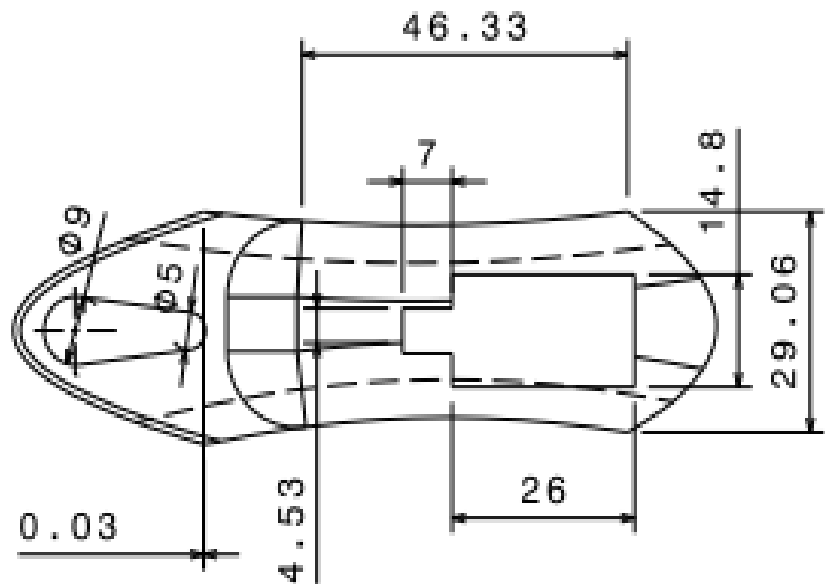
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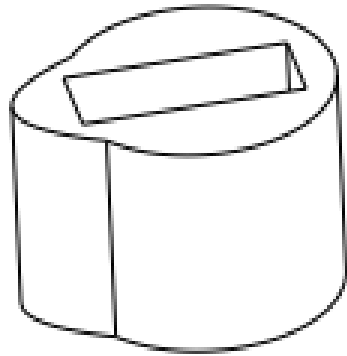
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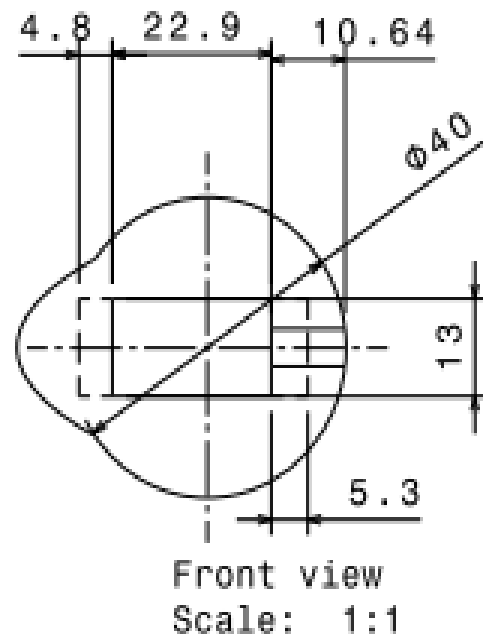
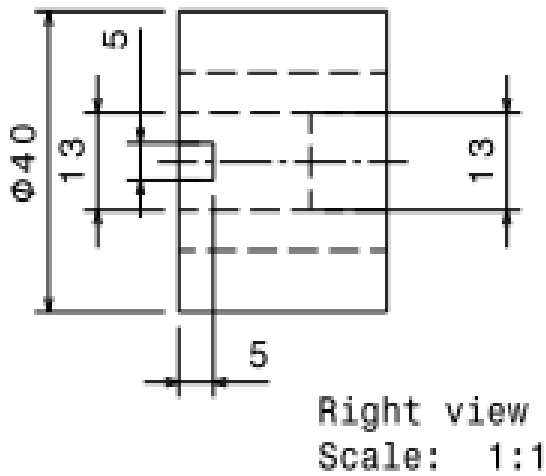
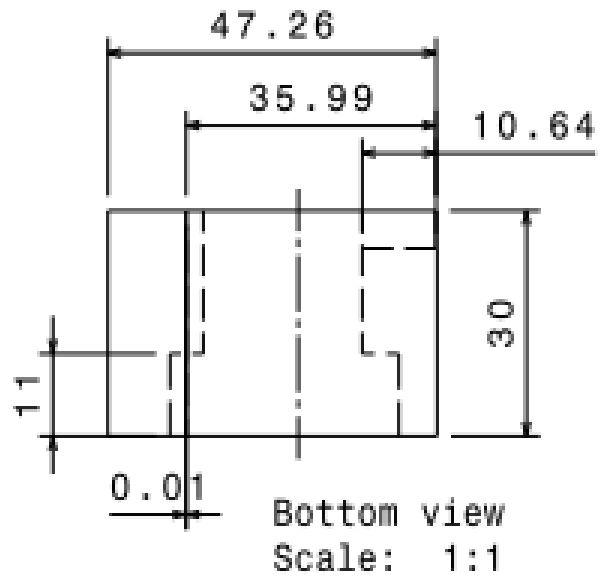
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Isometric view
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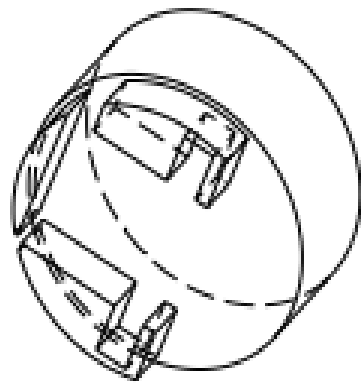
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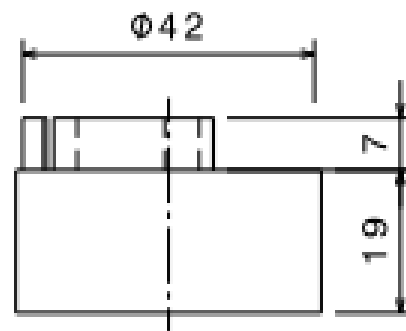
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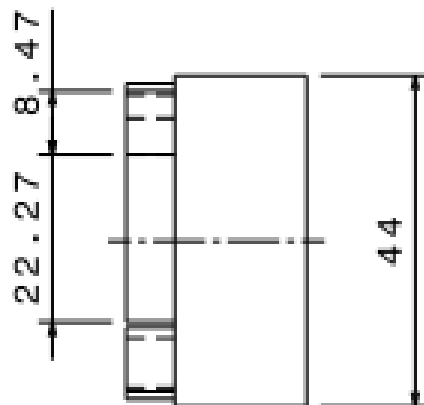
I	-
H	-
G	-
F	-
E	-
D	-
C	-
B	-
A	-



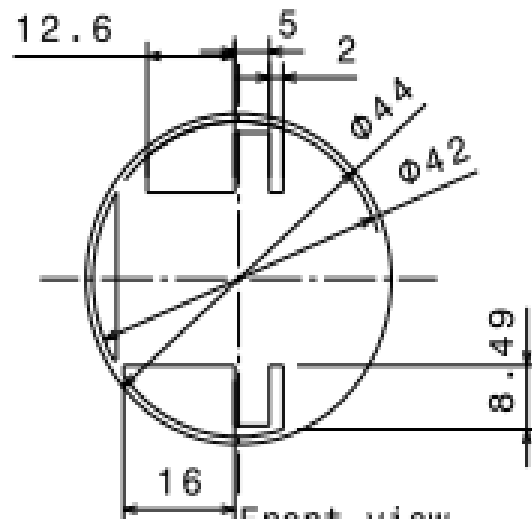
Isometric view
Scale: 1:1



Bottom view
Scale: 1:1



Right view
Scale: 1:1



Front view
Scale: 1:1

DESIGNED BY:
Intissar Issaoui
DATE:
5/12/2024

CHECKED BY:
XXX
DATE:
XXX

SIZE
A4



SCALE
1:1

WEIGHT (kg)
0.03

DRAWING NUMBER
Upper Base

SHEET
1/1

I	-
H	-
G	-
F	-
E	-
D	-
C	-
B	-
A	-

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Résumé

L'intégration de l'intelligence artificielle (IA) dans le contrôle des bras robotiques offre des solutions innovantes pour améliorer leurs performances. En utilisant l'apprentissage par renforcement (RL), l'objectif est d'automatiser les mouvements précis des bras dans des environnements dynamiques. L'étude détaillée de la conception et de la construction des bras robotiques joue un rôle crucial pour mettre en lumière le potentiel élevé de cette combinaison IA-robotique dans la transformation des industries et l'amélioration de notre quotidien.

Mots clés : Intelligence Artificielle, Bras Robotique, Performance, RL, Conception.

Abstract

The integration of artificial intelligence (AI) into the control of robotic arms offers innovative solutions to enhance their performance. By utilizing reinforcement learning (RL), the aim is to automate precise arm movements in dynamic environments. A detailed study of the design and construction of robotic arms plays a crucial role in highlighting the high potential of this AI-robotics combination in transforming industries and improving our daily lives.

Keywords: Artificial Intelligence, Robotic Arm, Performance, RL, Design

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